

model that most advisors and investors have followed since. His Nobel Prize winning effort created *modern portfolio theory* (MPT), which provides for the construction of investment portfolios that maximize expected returns based upon a targeted level of risk. His efforts showed that higher returns are achieved by taking on higher risk, while also recognizing what he called an *efficient frontier*, which defines the maximum possible expected return for a given level of risk.

The key for any investor employing MPT is to explicitly consider risk. While risk is not a palatable thought for retail investors—many of whom prefer to dream of risk-free million-dollar returns—there can be no reward without risk. The Securities and Exchange Commission, which regulates securities markets in the United States, has this advice about risk for investors:

When it comes to investing, risk and reward are inextricably entwined. You’ve probably heard the phrase “no pain, no gain.” Those words come close to summing up the relationship between risk and reward. Don’t let anyone tell you otherwise. All investments involve some degree of risk. If you intend to purchase securities—such as stocks, bonds, or mutual funds—it’s important that you understand before you invest that you could lose some or all of your money. The reward for taking on risk is the potential for a greater investment return.³

We’ll tackle the specifics of quantifying risk shortly, mainly through a discussion of volatility. Similarly, we will dive into how to approach absolute returns and the returns per unit of volatility, or risk-reward ratio.